ChatGPT - 5.2

# User Activity Monitoring System

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Source: https://chatgpt.com/c/69a1236a-456c-8324-ac0d-d1e760d33d80

i want to add one more component to my project where i can check the system is idel, or check the user away from keyboard or not also track the time ,or also duration,then this data will show on department hod and admin side and make the admin dashboard more improve like list of approved sign up ,unapproved , all show the preview what this syatem contain. . . . . .2

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Very good 👌 This is actually a **professional-level feature** — like employee monitoring + workflow analytics system.

Since you're already building a task workflow system (HOD → Department → Workman → SPOC → Admin), this feature will make your project much stronger 💪

Let's design it properly.

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## ✅ 1️⃣ New Component: User Activity Monitoring System

## 🎯 What You Want

You want to:

- ✅ Detect if system is **Idle**
- ✅ Detect if user is **Away from keyboard**
- ✅ Track:
  - Login time
  - Active time
  - Idle time
  - Total session duration
- ✅ Show this data to:
  - Department HOD
  - Admin

## 🔥 How Big Companies Do This

Companies like:

- 🏢 Zoho People
- 🏢 Hubstaff
- 🏢 Time Doctor

Track:

- Mouse movement
- Keyboard activity
- Browser focus
- App usage time
- Session logs

We will build a simplified but powerful version.

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## 🏗️ System Architecture

### 🖥️ Frontend (React / Next.js)

Track:

- Mouse movement
- Key press
- Window focus/blur
- Tab change
- Inactivity timer

### ⚙️ Backend (Flask / FastAPI)

Store:

- login_time
- last_activity_time
- idle_start_time
- total_idle_time
- total_active_time
- logout_time

### 🗄️ Database (PostgreSQL / MongoDB)

Create table:

```
user_activity_logs
--------------------
id
user_id
login_time
logout_time
total_active_time
total_idle_time
last_activity_time
status (Active / Idle / Offline)
```

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## 🧠 How to Detect Idle (Frontend Logic)

### Step 1: Detect User Activity

```
let idleTimer;
let idleTime = 0;
const IDLE_LIMIT = 60; // 60 seconds

function resetTimer() {
  idleTime = 0;
  sendStatus("active");
}

document.onmousemove = resetTimer;
document.onkeydown = resetTimer;

setInterval(() => {
  idleTime++;
  if (idleTime >= IDLE_LIMIT) {
    sendStatus("idle");
  }
}, 1000);
```

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## ⏱️ What Data Should Be Sent to Backend?

Every 30 seconds:

```
{
  "user_id": 101,
  "status": "active",
  "active_duration": 120,
  "idle_duration": 30
}
```

Backend will calculate total duration.

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## 📊 What HOD & Admin Should See

Now this is important — dashboard must be powerful.

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## 👨‍💼 HOD Dashboard View

HOD can see:

- List of department users

- Current status:
  - 🟢 Active
  - 🟡 Idle
  - 🔴 Offline
- Today's:
  - Total login time
  - Active time
  - Idle time
  - Tasks completed
- Productivity %

Example table:

| User | Status | Active Time | Idle Time | Tasks Done | Productivity |
|---|---|---|---|---|---|

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## 👑 Admin Dashboard (Advanced Version)

Admin can see:

### 📌 1. User Management Section

- Approved Signups
- Unapproved Signups
- Rejected Users
- Role (Admin / HOD / Workman / SPOC)

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### 📌 2. Live System Overview

- Total users online
- Total idle users
- Total offline users
- Total tasks in system
- Total completed tasks

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### 📌 3. Activity Analytics

- Daily login chart
- Department productivity chart
- Most idle department

- Most productive user

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### 📌 4. System Preview Section

Admin should see:

- Total departments
- Total employees
- Active workflows
- Pending approvals
- Task transfer chain visualization

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## 🎨 Make Admin Dashboard Look Professional

Add:

- Cards (Top statistics)
- Graphs (Bar + Line)
- Filters:
  - By department
  - By date
  - By user
- Search bar

Use:

- Recharts
- Chart.js
- ShadCN UI
- Tailwind cards

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## 🏆 Extra Powerful Features (Optional)

If you want next-level project:

- Screenshot capture every 10 min
- Auto logout after 15 min idle
- Browser tab tracking
- Export activity report (PDF / Excel)
- Attendance auto calculation

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## ⚠️ Important Legal Note

If this system is for real company use:

You MUST:

- Inform users activity is tracked
- Add privacy policy
- Add consent checkbox

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## 💡 Since You Are B.Tech Student

This feature will:

- Make your project industry-level
- Very impressive in placements
- Good for analytics career also
- Can be extended into SaaS product

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## 🚀 If You Want

I can now give you:

- Complete database schema
- Full Flask API structure
- Role-based access logic
- React dashboard layout design
- Productivity formula logic
- OR full system architecture diagram

Tell me your backend:
👉 Flask or FastAPI?

And database:
👉 MongoDB or PostgreSQL?